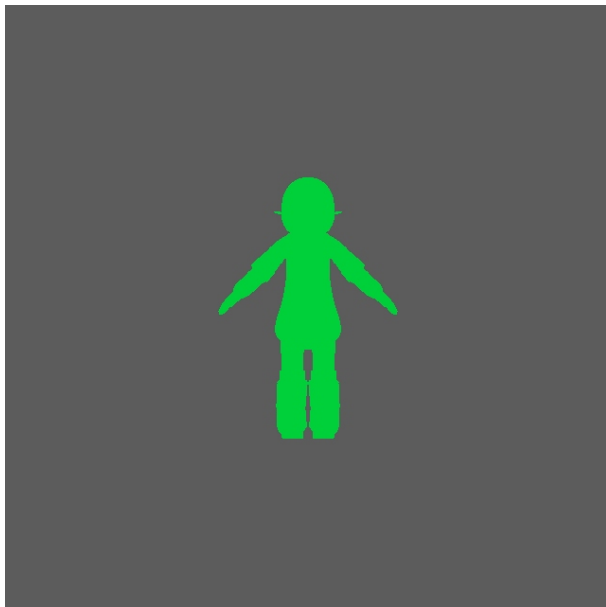


Nini. Modeling QC Feedback

Asset	Nini
Reviewed file	D:/RIGS/Nini/AUR_Nini_Modeling_v0001.fbx (58 meshes, 44,482 vertices)
Body mesh used for symmetry and limb tests	c_body_GEO
Reference	Modeling QC Checklist (Topology/Model design checks, Tech checks)
Reviewed by	Rigging department (automated Model QC v002)
Date	16 September 2026
Status	NOT APPROVED. Fix the items in section 2 and resubmit. Items in section 3 are recommended.

1. Summary

7 blocking issues, 6 recommendations, 11 checks passed. Blocking issues must be fixed before rigging starts.



Front view.



Side view. The character should face +Z.

2. Must fix (blockers)

2.1 N-gons on deforming meshes

c_body_GEO: 12 n-gons near [[0.6, 110.8, 11.0], [0.7, 113.8, -6.7], [0.4, 112.1, 11.0]]; c_hood_GEO1: 2 n-gons near [[2.5, 122.8, -11.5], [0.0, 122.8, -11.9]]; c_pants_GEO: 2 n-gons near [[0.9, 64.6, -11.4], [-0.0, 64.6, -11.3]]. Rebuild as quads.

2.2 Vertices in the same position (unmerged seams)

c_body_GEO: 5 pairs near [[0.0, 113.851, -6.701], [0.0, 110.322, 10.458], [0.0, 112.019, 11.032]]; c_button_GEO: 2 pairs near [[-1.57, 105.4, 7.841], [0.916, 105.056, 7.37]]; c_pants_GEO: 1 pairs near [[-0.0, 66.845, -10.62]]. Merge the vertices (usually the mirror seam).

2.3 Transforms not frozen, pivots not at the origin

Unfrozen: R_specular_GEO t=[-0.3, 0.01, -0.07] r=[0.0, 0.0, 0.0] s=[1.0, 1.0, 1.0]; c_eyelashes_GEO t=[-1.82, -49.78, -6.91] r=[5.25, -13.05, -1.52] s=[0.419, 0.419, 0.419]; c_furBalls_GEO t=[0.0, -0.87, 1.97] r=[0.0, 0.0, 0.0] s=[1.0, 1.0, 1.0]; c_placeholder_bootBase t=[31.85, 0.07, 0.0] r=[0.0, 0.0, 0.0] s=[1.0, 1.125, 1.0]; nini_GRP (group) t=[0.0, 0.0, 2.0] r=[0.0, 0.0, 0.0] s=[1.0, 1.0, 1.0]; c_hair_GEO (group) t=[0.0, 0.0, -2.0] r=[0.0, 0.0, 0.0] s=[1.0, 1.0, 1.0].

1.0]; c_eyes_GRP (group) t=[4.88, 119.78, 4.84] r=[-5.03, 13.13, 0.34] s=[2.386, 2.386, 2.386]; teeth_top (group) t=[0.0, 0.22, -0.5] r=[0.0, 0.0, 0.0] s=[1.163, 1.102, 1.102]; teeth_low (group) t=[-0.0, -1.33, -0.77] r=[175.63, 180.34, -0.45] s=[1.163, 1.317, 1.102]. Pivot not at the origin: 58 meshes (e.g. C_tongue_GEO, L_lowCanine_GEO, L_lowFront_GEO, L_lowIncisor_GEO). Freeze translate, rotate and scale and put every pivot at (0, 0, 0).

2.4 Naming convention

Not matching side_name_GEO / side_name_GRP: c_hood_GEO1, c_placeholder_bootBase, c_hair_GEO, teeth_top, teeth_low. Mixed-case side prefixes (C, L, R, c, l, nini, r, teeth): pick one case for the whole asset.

2.5 Locked vertex normals

58 of 58 meshes (C_tongue_GEO, L_lowCanine_GEO, L_lowFront_GEO, L_lowIncisor_GEO, L_lowMolar1_GEO...). Locked normals do not update on deformation. Unlock them (Mesh Display > Unlock Normals). FBX export locks normals; deliver a Maya scene.

2.6 Delivery format

Only an FBX was delivered. Deliver a Maya ASCII (.ma) scene.

2.7 Limb plane: arm

Middle joint keeps [0.95, 3.99] deg on the minor axes; end guide drifts 1.59 cm (limb radius 1.2 cm). Realign the arm so the bend happens in one plane.

3. Should fix (recommended)

3.1 Meshes under 2000 vertices

54 of 58 meshes. Deformers on these run on the CPU. Combine meshes that deform together (e.g. C_tongue_GEO, L_lowCanine_GEO, L_lowFront_GEO, L_lowIncisor_GEO, L_lowMolar1_GEO).

3.2 Per-face material assignment

L_pupil_GEO: 3 shading groups; R_pupil_GEO: 3 shading groups; 6 groupId nodes. One material per mesh unless lookdev needs the split.

3.3 Poles near deformation areas

elbow: 4 poles within 7 cm (valence, x, y, z: [[5, 22.4, 84.3, -4.9], [5, 25.8, 87.6, -5.4]]); shoulder: 7 poles within 7 cm (valence, x, y, z: [[3, 11.6, 102.1, -5.8], [5, 13.9, 98.5, -7.1]]); wrist: 71 poles within 7 cm (valence, x, y, z: [[5, 42.3, 69.0, -1.5], [5, 42.5, 68.9, 0.0]]). Keep poles off the fold loops.

3.4 Limb plane alignment: leg

Passes with a small deviation: [0.45, 2.2] deg on the minor axes, end guide drifts 1.00 cm (limb radius 2.3 cm). Optional: move the leg so the three landmarks share one plane.

3.5 Pre-bend: leg

22.6 degrees. 5 to 10 degrees is enough for the IK solver.

3.6 Pre-bend: arm

18.0 degrees. 5 to 10 degrees is enough for the IK solver.

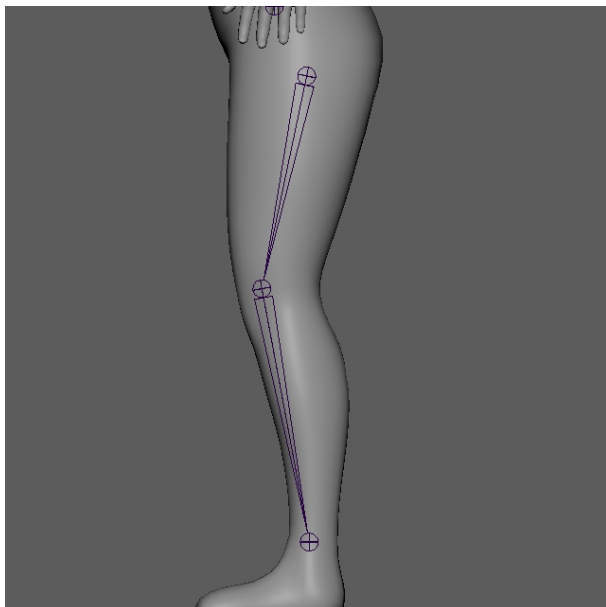
4. Limb plane check (single rotation axis)

Method: a 3-joint chain is placed on each limb (hip, knee, ankle and shoulder, elbow, wrist) on the landmark locators. The chain is oriented with Orient Joint (primary X down the chain, secondary axis toward the bending plane). A limb is in one plane when the middle joint carries rotation on one axis only. The root can have any orientation. The two minor axes of the middle joint are then set to 0 and the drift of the end guide is measured. Pass: the end guide stays approximately in place, inside the limb volume and close to its center.

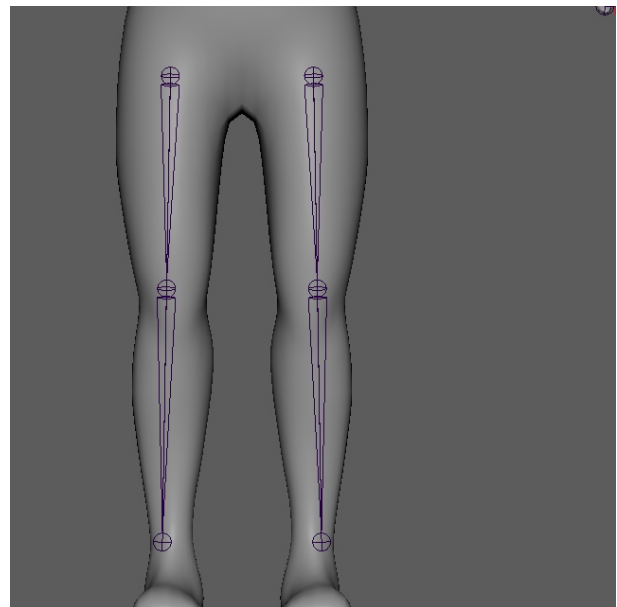
Chain	Middle joint orient (X, Y, Z) after Orient Joint	Orient with the other axes at 0	Residual removed	End guide drift after zeroing	Pass
Leg	knee: -0.45, 22.48, -2.2	knee: 0.0, 22.48, 0.0 (Y only)	0.45 deg, 2.20 deg	ankle moves 1.00 cm ([1.0, -0.0, 0.0]); ankle radius about 2.3 cm	YES, small deviation
Arm	elbow: 0.95, -3.99, -17.6	elbow: 0.0, 0.0, -17.6 (Z only)	0.95 deg, 3.99 deg	wrist moves 1.59 cm ([-1.15, -1.1, 0.01]); wrist radius about 1.2 cm	NO

Leg: bend axis Y, pre-bend 22.6 degrees, pole direction [0.09, -0.01, 1.0] (offset 5.04 cm), plane normal [0.996, -0.002, -0.094], secondary axis xup. Verdict: PASS, small deviation.

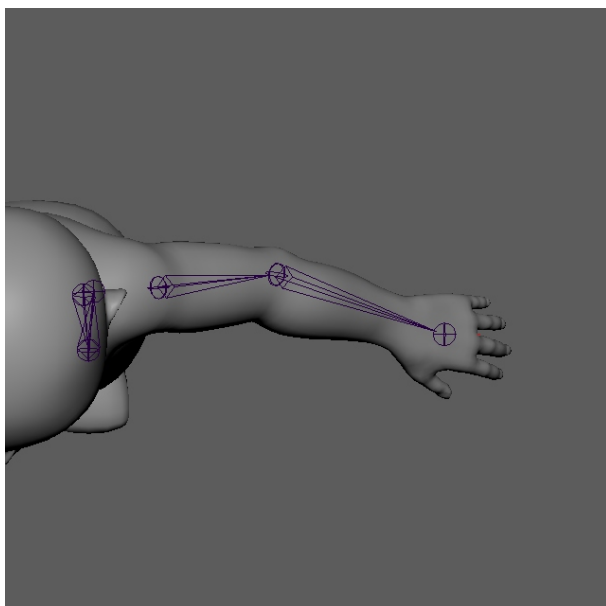
Arm: bend axis Z, pre-bend 18.0 degrees, pole direction [-0.09, -0.23, -0.97] (offset 2.92 cm), plane normal [-0.717, -0.66, 0.224], secondary axis zdown. Verdict: FAIL.



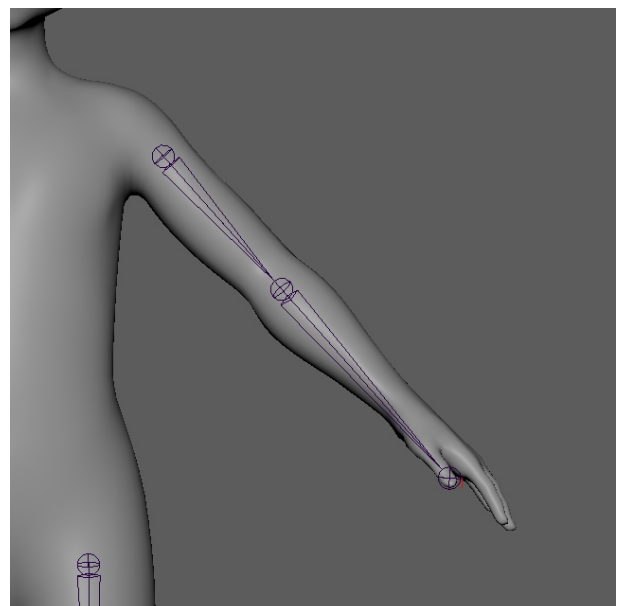
Leg chain, side view. Red locator: ankle landmark.



Leg chain, front view, after zeroing the minor axes.



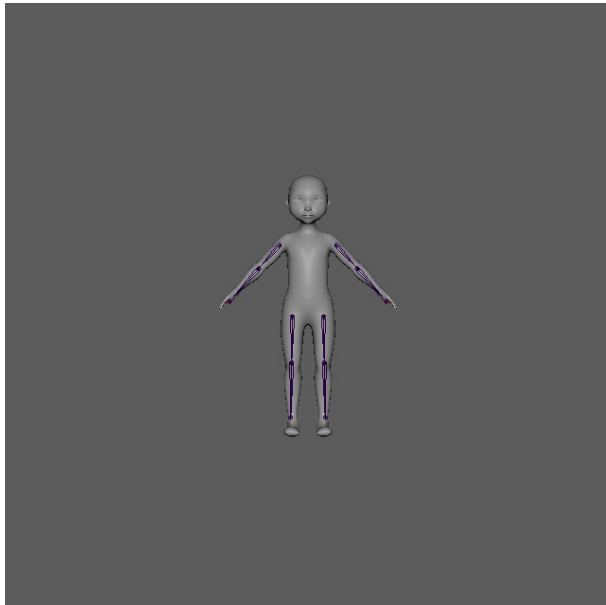
Arm chain, top view. Red locator: wrist landmark.



Arm chain, front view, after zeroing the minor axes.

5. Checks passed

Check	Result
Non-manifold, lamina faces	None on any mesh.
Construction history	None.
Vertex tweaks (pnts)	All zero.
Duplicate names	None.
Namespaces, unknown nodes	None.
One shape per transform, no mesh under a mesh	OK.
Body symmetry	5870 vertices per side, 222 on the center line, max deviation 0.000 cm.
Triangles	None on deforming meshes.
Ground contact	Lowest point at Y = 0.054.
Unused shaders, extra cameras, layers, sets	None (37 shading groups, all assigned).
Neutral expression, feet direction, cloth/body elbow match	Visual check by the reviewer. See screenshots.



Body mesh with the guide chains.



Face, expression check.

6. Per-mesh summary

Mesh	Verts	Flags
C_tongue_GEO	235	pivot off origin, locked normals
L_lowCanine_GEO	182	pivot off origin, locked normals
L_lowFront_GEO	138	pivot off origin, locked normals
L_lowIncisor_GEO	182	pivot off origin, locked normals
L_lowMolar1_GEO	114	pivot off origin, locked normals
L_lowMolar2_GEO	114	pivot off origin, locked normals
L_lowMolar3_GEO	114	pivot off origin, locked normals
L_pupil_GEO	898	32 tris, pivot off origin, locked normals, 3 materials
L_sclera_GEO	1042	32 tris, pivot off origin, locked normals
L_specular_GEO	21	20 tris, pivot off origin, locked normals
L_topCanine_GEO	182	pivot off origin, locked normals
L_topFront_GEO	182	pivot off origin, locked normals

Mesh	Verts	Flags
L_topIncisor_GEO	182	pivot off origin, locked normals
L_topMolar1_GEO	114	pivot off origin, locked normals
L_topMolar2_GEO	114	pivot off origin, locked normals
L_topMolar3_GEO	114	pivot off origin, locked normals
R_lowCanine_GEO	182	pivot off origin, locked normals
R_lowFront_GEO	138	pivot off origin, locked normals
R_lowIncisor_GEO	182	pivot off origin, locked normals
R_lowMolar1_GEO	114	pivot off origin, locked normals
R_lowMolar2_GEO	114	pivot off origin, locked normals
R_lowMolar3_GEO	114	pivot off origin, locked normals
R_pupil_GEO	898	32 tris, pivot off origin, locked normals, 3 materials
R_sclera_GEO	1042	32 tris, pivot off origin, locked normals
R_specular_GEO	21	20 tris, unfrozen, pivot off origin, locked normals
R_topCanine_GEO	182	pivot off origin, locked normals
R_topFront_GEO	182	pivot off origin, locked normals
R_topIncisor_GEO	182	pivot off origin, locked normals
R_topMolar1_GEO	114	pivot off origin, locked normals
R_topMolar2_GEO	114	pivot off origin, locked normals
R_topMolar3_GEO	114	pivot off origin, locked normals
c_armcuffs_GEO	504	pivot off origin, locked normals
c_bangs_GEO	1254	pivot off origin, locked normals
c_body_GEO	11962	12 n-gons, 5 duplicate verts, pivot off origin, locked normals
c_bootCuffs_GEO	520	pivot off origin, locked normals
c_bottomsSleeves_GEO	720	pivot off origin, locked normals
c_braids_GEO	4656	pivot off origin, locked normals
c_button_GEO	794	2 duplicate verts, pivot off origin, locked normals
c_circle_GEO	25	pivot off origin, locked normals
c_collar_GEO	794	pivot off origin, locked normals
c_embroider_GEO	231	pivot off origin, locked normals
c_ends_GEO	146	pivot off origin, locked normals
c_eyelashes_GEO	108	unfrozen, pivot off origin, locked normals
c_furBalls_GEO	196	unfrozen, pivot off origin, locked normals
c_furCuffs_GEO	108	pivot off origin, locked normals
c_gloves_GEO	1388	pivot off origin, locked normals
c_hairties_GEO	800	pivot off origin, locked normals
c_hood_GEO1	1155	2 n-gons, pivot off origin, name, locked normals
c_legwarmers_GEO	1860	pivot off origin, locked normals
c_lowGums_GEO	986	pivot off origin, locked normals
c_pants_GEO	1530	2 n-gons, 1 duplicate verts, pivot off origin, locked normals
c_placeholder_bootBase	2400	unfrozen, pivot off origin, name, locked normals
c_ribbons_GEO	900	pivot off origin, locked normals
c_sidePieces_GEO	304	pivot off origin, locked normals
c_topGums_GEO	1209	pivot off origin, locked normals
c_tunic_GEO	2227	pivot off origin, locked normals
l_eyebrow_GEO	42	pivot off origin, locked normals

Mesh	Verts	Flags
r_eyebrow_GEO	42	pivot off origin, locked normals

Resubmission: deliver the asset as a Maya ASCII scene with section 2 fixed. Rigging will re-run this check before starting the rig.